Radio Frequency Interference Signature Detection in Radar Remote Sensing Image Using Semantic Cognition Enhancement Network

Mingliang Tao, *Member, IEEE*, Jieshuang Li, Junli Chen, Yanyang Liu, Yifei Fan, Jia Su, *Member, IEEE*, and Ling Wang

***Abstract*— Radio frequency interference (RFI) is a significant threat to accurate microwave remote sensing. The RFI signals manifest themselves in unpredictable locations and patterns in the image, which will cause measurement distortion and image degradation or even lead to wrong retrievals of the geophysical parameters. Accurate detection of RFI artifacts is a prerequisite step to preserve the overall quality of remote sensing quality. In this article, a semantic cognitive enhancement network for RFI signature detection is proposed. It employs an encoder–decoder architecture, which incorporates the atrous spatial pyramid pooling, depthwise convolution, and self-attentional mechanism. Rather than detecting the existence of RFI artifacts for an entire image, the proposed scheme can realize RFI recognition in a pixelwise manner without setting predefined thresholds. Extensive experimental results on diverse scenarios in Sentinel-1 images with various RFI types are provided, which demonstrates robust detection performance for both strong and weak interference without requiring a large number of training samples.**

***Index Terms*— Interference detection, radio frequency interference (RFI), semantic cognition network, Sentinel-1, synthetic aperture radar (SAR).**

I. Introduction

A. Background and Motivation

SYNTHETIC aperture radar (SAR) is an active remote sensing instrument that can provide information essential for understanding the Earth's environment and its evolution [1]. As an Earth exploration-satellite service, the operating frequency band of SAR satellites is strictly protected by spectrum management and regulations from the International Telecommunication Union (ITU) [2]. However, it is still unavoidable to receive spurious signals emitted by other radio services working in the same or adjacent frequency band, which is referred to as radio frequency interference (RFI).

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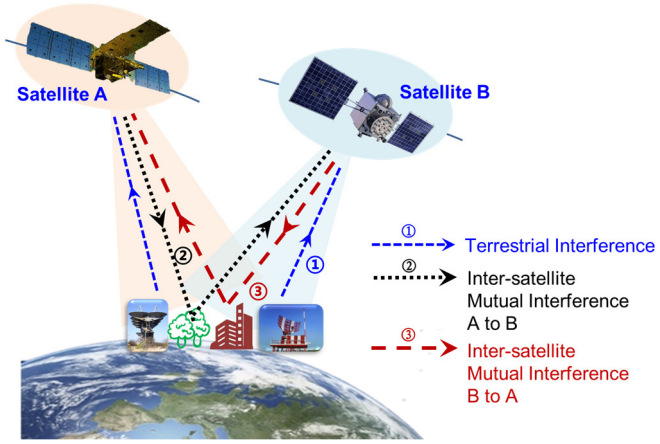


Fig. 1. Illustration of potential RFI sources and interfering mechanism for a spaceborne SAR system.

In recent years, with the free open access to the European Sentinel-1 data, more and more RFI instances in Sentinel-1 data have been reported [3]. Fig. 1 shows the potential RFI sources that affect a spaceborne SAR system. The well-documented RFI sources are originating from the terrestrial radars and radiolocation services [4] and also the mutual spaceborne interference scattered from other satellites that share a common spectrum and an overlapping illumination area [5], [6]. The extension of radio local area network (RLAN) services could hasten the likelihood of interference to the current C-band SAR system [7]. RFI pollution may distort the amplitude and phase correlation of the raw echo and adversely affect the accuracy of the imaging and interpretation process. Therefore, to maintain the overall data quality, it is very necessary to identify or mark the exact part of radar data affected by RFI, rather than discard the RFI-contaminated observations directly [8].

B. Related Works

Over the past years, a variety of RFI detection and mitigation techniques have been proposed to alleviate the adverse impact. A thorough overview of the status and trend of RFI detection and mitigation techniques are given in [9] and [10]. RFI is typically identified by distinguishing from the expected target echo return by their temporal, spectral, spatial,

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polarimetry, or statistical properties. From the system design level, Monti-Guarnieri *et al.* [3] proposed the monitoring of rank echoes of the Sentinel-1 system during the echo acquisition phase and derived a global probability map of RFI in C-band after a long-time continuous analysis. In addition to the data acquisition phase, the majority of the existing methods also postprocess the acquired echo data to detect RFI for Level-0 raw data, ranging from the time domain [11], [12], frequency domain [13], azimuth-frequency domain [14], range-azimuth-space domain [15], or time–frequency domain [16]. However, raw echo is not always available for remote sensing end users. Interference detection applied directly in the image domain is less computationally intensive. Leng *et al.* [4] distinguished the difference in the backscatter characteristics between the RFI and the target under different polarizations for detection. Yang *et al.* [17] and Huang *et al.* [18] proposed a subspace filtering method for single-look complex (SLC) images. Chojka *et al.* [19] used image processing methods to flag images with RFI artifacts. Then, the flagged images are removed from the subsequent processing workflow. Instead of performing the RFI mitigation for a single image, an RFI extraction and mitigation method is proposed in [20] by exploiting the coupling correlation among multiple time-series images. However, the aforementioned model-based methods are limited to finding a general adaptive static to adjust the variation of land cover and RFI signatures.

In recent years, due to the versatility of deep convolution neural networks and the accumulation of remote sensing big data, data-based intelligent learning and processing in remote sensing images have become a noneligible trend [21]. A neural network could reveal hidden multiscale features that benefit processing, and its feasibility on raw echo was demonstrated in [21]–[23]. Alternatively, from the perspective of image processing, Artiemjew *et al.* [24] applied the LeNet convolution filter to remove the RFI-affected data that affect interferometric processing. Furthermore, Saini *et al.* [25] used the UNET-Like convolutional network to segment the whole interference area on the Sentinel-1 image but only produces a general RFI mask without extracting the patterns of the RFI. This idea is further extended to the time–frequency spectrogram of the raw echo [26]. Yan *et al.* [27] used atrous convolution in the U-Net architecture to replace the convolution kernel of the coding layer and decoding layer to segment RFI in the time–frequency domain. This operation is intended to improve the network's supplement to detail information by increasing the receptive field to obtain more context information.

Although exiting data-based methods have shown promising results, a critical challenge is that the training database should include diverse RFI cases to guarantee performance. Meanwhile, these networks are only capable of dealing with relatively homogeneous scenarios with strong interference, and their generalization capability to weak interference is limited to some extent.

C. Contributions

To further improve the detection performance in various RFI scenarios, our efforts focus on investigating advanced data-driven deep learning methods to realize effective and efficient RFI flagging. The major contributions are summarized as follows.

1) A semantic cognitive enhancement network for interference detection on a single Sentinel-1 Level-1 image is proposed by employing the fusion of atrous spatial pyramid pooling (ASPP), depthwise convolution, and self-attentional mechanism. It effectively transforms the traditional binary hypothesis detection problem into an image semantic segmentation problem.

2) It is capable of distinguishing the RFI signatures in a pixelwise manner for amplitude-only images while without requiring a large volume of the training dataset. The extracted RFI patterns could be utilized for further tracing back analysis of the RFI source reporting.

3) Extensive experimental results using real measured Sentinel-1 image data are provided. The selected testing scenarios are diverse with different types of RFI signatures. The performance comparison with several state-of-the-art (SOTA) techniques verifies that the proposed method can flag out the RFI artifacts with high accuracy and efficiency, even in the scenarios with weak interference and complex distributed interference.

The rest of this article is organized as follows. The RFI characteristics and the detection problem are formulated in Section II. Section III introduces the theory and methodology of the proposed method in detail. Section IV provides extensive experimental results of real measured Sentinel-1 image datasets for representative scenarios, and the concluding remarks are provided in Section V.

II. Problem Formulation

Fig. 2 compares the images acquired for various areas by the European Sentinel-1 system at different acquisition passes. The first row corresponds to the RFI-contaminated images, while the second row represents the images without interference. In comparison with the same scenario in the vertical column, it is shown that the interference is distributed over land or sea and presents unique radiometric characteristics differing from the land covers. Fig. 2(a) and (b) shows the artifacts resulting from typical terrestrial RFI sources such as ship-based radiolocation radars [20]. The consecutive rectangular blocks clearly show the pulsed nature of the interfering signals, which are highly related to the waveform characteristics, such as power, pulsewidth, and pulse repetition frequency.

Unlike the terrestrial RFI, the RFI signatures in Fig. 2(c) and (d) present different signatures. The bright strips in Fig. 2(c) as well as the regular patterns in Fig. 2(d) are a result of long-duration intersatellite mutual interference between space-based radars. It is a kind of scatter-wave RFI modulated by the illuminated land covers, which is well studied in multiple representative cases [5]. It covers a great portion of the image, which indicates that the duration of the interfering process is longer than that of the terrestrial RFI sources. Both types of interferences would degrade image quality and affect subsequent image interpretation. Therefore, it is of great necessity to effectively detect and localize the interference.

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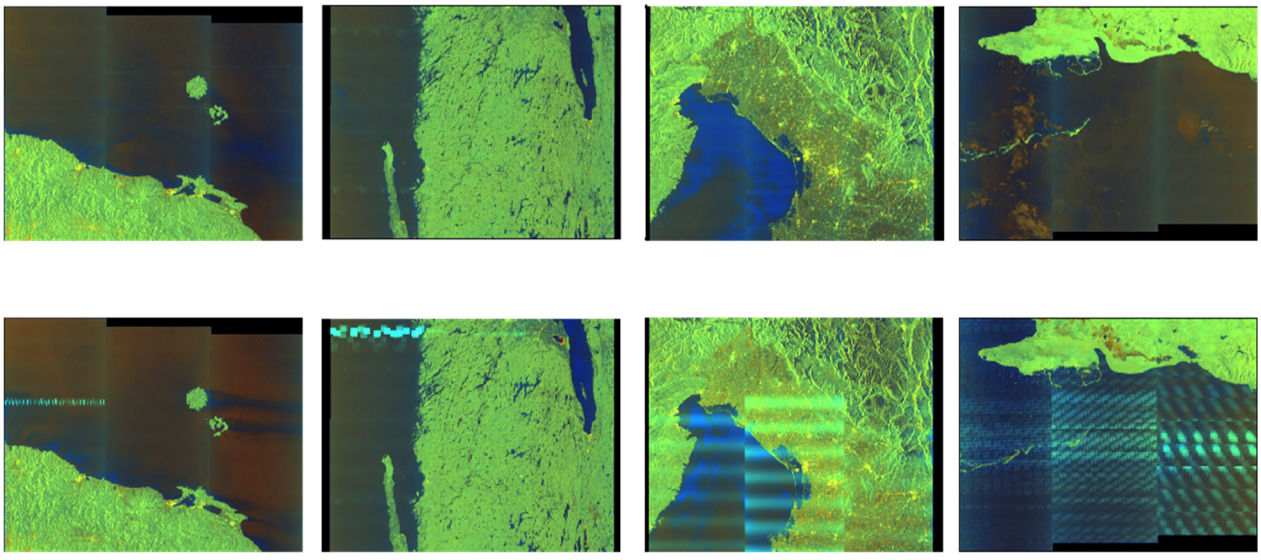


Fig. 2. Comparisons of SAR images with and without RFI artifacts. (a) and (b) Terrestrial interference. (c) and (d) Intersatellite mutual interference.

The RFI detection problem could be realized on two levels. The first level is to detect the existence of RFI for the whole image without specifying the exact portions, which is relatively easier to realize and commonly applied for mass image processing applications [8], [25]. A more advanced detection scheme is realized in a pixelwise manner to accurately locate the RFI artifacts. Generally, the RFI detection problem could be formulated as a problem of binary hypothesis testing, i.e.,

$$\begin{cases} H_0 : \mathbf{X} = \mathbf{S} + \mathbf{N}, & T < \eta \\ H_1 : \mathbf{X} = \mathbf{S} + \mathbf{I} + \mathbf{N}, & T \geq \eta \end{cases} \tag{1}$$

where H_0 represents the null hypothesis that the original image is without RFI and H_1 represents the alternative hypothesis that the image is RFI-polluted. $\mathbf{X}$ denotes the measurements, $\mathbf{S}$ denotes the echoes of illuminated land covers, $\mathbf{N}$ denotes the noise, and $\mathbf{I}$ represents the RFI artifacts. T is the detection statistic and η is the critical value for hypothesis testing.

To solve (1), the model-based methods are required to give a threshold or static for hypothesis testing. However, the diverse variety of land covers and RFI signatures makes it rather challenging to find a general static to effectively detect and localize the interference with great efficiency.

III. Theory and Methodology

In this article, instead of managing to find an optimal static for each pixel to discriminate the RFI artifacts, we propose an alternative data-driven scheme by utilizing the contextual information among pixels. The original detection problem is converted into a semantic segmentation problem for each pixel. In semantic segmentation, all pixels corresponding to RFI are flagged out using one class label.

For an input image, semantic segmentation is realized by distinguishing two regions g_1 and g_2 of the image that has a particular meaning, which could be expressed as

$$\begin{cases} g_1(\mathbf{S};\mathbf{N}) \cup g_2(\mathbf{S};\mathbf{I};\mathbf{N}) = \mathbf{X} \\ g_1(\mathbf{S};\mathbf{N}) \cap g_2(\mathbf{S};\mathbf{I};\mathbf{N}) = \phi \end{cases} \tag{2}$$

where ϕ denotes the empty set that the regions g_1 and g_2 are nonintersecting with each other and each region satisfies region-specific connectivity. If the connectivity constraint is removed, then the division of the set of pixels is called pixel classification, and each set of pixels is called a class. Therefore, the original segmentation problem could be regarded as a pixelwise classification problem.

To solve this problem, we proposed a semantic cognitive enhancement network in this article. As shown in Fig. 3, it employs a classical encoder–decoder structure. By applying dilated convolution on multiple scales, the encoder part deals with multiscale context information. It also generates low-resolution feature maps that are suitable for identification. The decoding module samples the feature representation upward into a full-resolution segmentation image with boundary refining.

Consider an input image with a size of $M \times N \times 3$, and then, the output feature map after convolutional layers could be expressed as

$$O_j^{(l)}(m,n) = \sum_{i-1}^{M} \sum_{u,v=0}^{S-1} W_{ji}^{(l)}(u,v) \cdot L_i^{(l-1)}(m-u, n-v) + b_j^{(l)} \tag{3}$$

$$L_j^{(l)}(m,n) = f\left(O_j^{(l)}(m,n)\right) \tag{4}$$

where $f(\cdot)$ denotes the nonlinear activation operator, $O_j^{(l)}(m,n)$ denotes the output feature map, and $W_{(\cdot)}$ and $b_{(\cdot)}$ are the network weights and bias, respectively.

The output of the final layer is a flagging mask with the size of $M \times N \times 1$, with each pixel on the channel having

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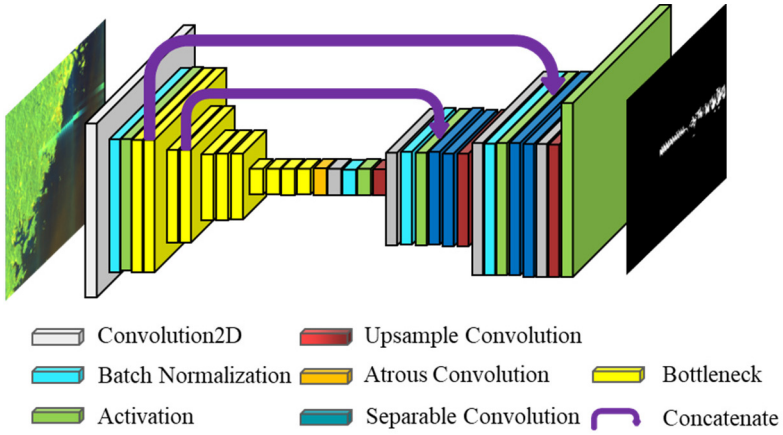


Fig. 3. Architecture of the proposed semantic cognition enhanced network.

a different score, which is mapped by the softmax activation function to a final 2-D segmentation map, which can be defined as

$$\boldsymbol{\sigma}(\mathbf{z})_i = \frac{e^{z_i}}{\sum_{c=1}^{C} e^{z_c}} \tag{5}$$

where i is the total number of neurons in the last layer and p_i is the output probability. It applies the standard exponential function to each element of the input vector $\mathbf{z}$ and normalizes these values by dividing by the sum of all these exponentials.

The output results measure the statistical difference between the segmentation result and the label according to the pixelwise cross-entropy loss function, which is defined as [28]

$$\begin{aligned}\text{Loss} = -\frac{1}{MN}\sum L(m,n)\log\left(1-\hat{L}(m,n)\right) \\ -(1-L(m,n))\log\left(1-\hat{L}(m,n)\right)\end{aligned} \tag{6}$$

where $\hat{L}(m,n)$ represents the predicted values.

The values of W and b are adjusted by backpropagating the network so that the network segmentation results keep approximating the labeled results, i.e.,

$$\frac{\partial \text{Loss}}{\partial W_{ji}^{(l)}} = \sum_{m,n} \delta_j^{(l-1)}(m,n)\cdot \text{Loss}_j^{(l-1)}(m-u, y-v) \tag{7}$$

$$\frac{\partial \text{Loss}}{\partial b_j^{(l)}} = \sum_{m,n} \delta_j^{(l-1)}(m,n) \tag{8}$$

$$W_{ji}^{(l)} = W_{ji}^{(l-1)} + \gamma \cdot \frac{\partial \text{Loss}}{\partial W_{ji}^{(l-1)}} \tag{9}$$

$$b_{ji}^{(l-1)} = b_{ji}^{(l-1)} + \gamma \cdot \frac{\partial \text{Loss}}{\partial b_{ji}^{(l-1)}} \tag{10}$$

where γ is the learning rate during the optimization process.

In the following, we will introduce the definitions and functions of all modules used in the network architecture in Fig. 3, as well as the mathematical principles of the key modules affecting the final detection performance, and explain the rationale for designing these modules.

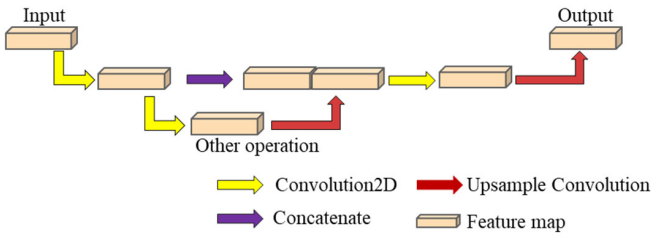


Fig. 4. Basic segmentation network model.

A. Basic Network Model

A basic segmentation network needs to extract image features through the convolution layer, as shown in Fig. 4. The Convolution2D (Conv2D) layer, batch normalization (BN) layer (dark cyan block), and activation layer (light green block) shown in Fig. 3 are considered to be the basic unit of neural network convolution layer [29]. Conv2D operation is used to extract the feature differences between interference and surrounding background, BN operation is used to prevent gradient explosion and gradient vanishing, and the nonlinear activation layer cuts the information after convolution mapping to enhance the learning ability of the network.

The whole feature extraction process is a downsampling process, so the network requires upsampling to restore the size of the output mask as identical to the original image. There are many existing upsampling methods, and the proposed method adopts bilinear interpolation as the upsample convolution (Upconv) layer. Given a set of sequences $[x_0, x_1, \ldots, x_k]$ and their corresponding values $[y_0, y_1, \ldots, y_k]$, the bilinear interpolation operation can be defined as

$$L(x) = \sum_{j=0}^{k} y_j \ell_j(x) \tag{11}$$

$$\ell_j(x) = \prod_{i=0, i\neq j}^{k} \frac{x - x_i}{x_j - x_i} \tag{12}$$

where $\ell_j(x)$ is the Lagrange interpolation basis function.

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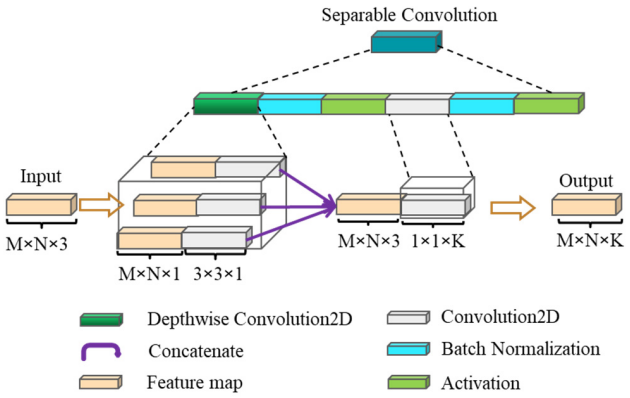


Fig. 5. Architecture of the separable convolution module.

A basic unit of convolution layer is usually implemented after the Upconv layer since the Upconv layer has no weight itself, which makes it impossible to adjust through the backpropagation. Finally, the information recovered after upsampling operation differs from the origin before the downsampling operation, and thus, the network uses the concatenate layer to supplement the detailed information.

B. Separable Conv2D

For remote sensing applications, it is rather difficult to obtain a sufficiently large number of labeled images for ground-truth training and validation. Oversized network architectures will cause overfitting under the condition of a small training dataset. The standard convolution layer of a neural network involves input × output × width × height parameters, where *width* and *height* are width and height of filter, respectively. For example, if an input channel of 3 and output of 3 are with 5 × 5 filters, this will have 225 parameters. Such a number of parameters would increase the possibility of overfitting. In depthwise convolution, it originated from the idea that the depth and spatial dimension of a filter can be separated. Different filters are convolved separately onto corresponding channels and then stacked thereafter. By doing so, only total $3 \times 5 \times 5 + 3 \times 3 = 84$ parameters are needed.

Therefore, we propose to use the depthwise convolution to achieve the network pruning [31]. The architecture of the proposed depthwise convolution module is shown in Fig. 5. In the depthwise convolution module, a single convolutional filter per each input channel is applied first. Then, a linear combination of the output is obtained by the pointwise convolution.

In the first step, each input channel is applied on a 3 × 3 depthwise convolution

$$\hat{O}_{k,l,n} = \sum_{i,j} \hat{K}_{i,j,m} \cdot F_{k+i-1,l+j-1,m} \tag{13}$$

where F is the input feature map.

For the second step, a 1 × 1 pointwise convolution $\tilde{K}$ is employed to produce a combination of the output

$$O_{k,l,n} = \sum_{m} \tilde{K}_{m,n} \cdot \hat{O}_{k-1,l-1,m}. \tag{14}$$

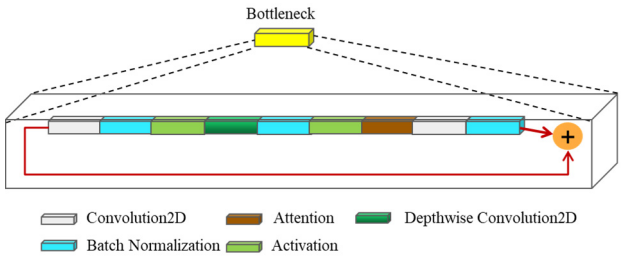


Fig. 6. Architecture of the bottleneck module.

In terms of generating new features, depthwise convolution and pointwise convolution play different roles. Spatial correlations are captured by the depthwise convolution, while the pointwise convolution is utilized for exploiting channelwise correlations.

C. Inverse Residual and BOTTLENECK

For small sample training datasets, a larger network architecture may cause overfitting, which leads to poor generalization capability and is difficult to deal with unknown interference scenarios. Therefore, a lightweight network would reduce the number of parameters of convolution layer operation and thus would help to improve the efficiency of the model. In the proposed network, the inverse residual, bottleneck layer, as well as separable convolution layer structures are used to replace the traditional convolution layer and realize the compression of the model. Fig. 6 shows the bottleneck module.

Compared with the residual module and the bottleneck residual module, this module realizes the compromise between accuracy and parameter quantity. The bottleneck residual module consists of totally three convolution layers of size 1 × 1, 3 × 3, and 1 × 1. The bottleneck layer first increases the dimension through 1 × 1 convolution layer, then goes through the depthwise convolution, and finally reduces the dimension through 1 × 1 convolution layer. The goal of this upsampling and downsampling operation is to enrich the amount of information during deep separable convolution processing and combine it with the residual structure.

D. Atrous Spatial Pyramid Pooling

As the main block of a deep neural network, the convolutional layer contains a set of filters, which slide over the height and width of the image and calculate the dot product between each element of the filter and the input at each spatial position. The convolutional filters with different sizes will produce various features. A larger sized kernel would introduce a larger perceptual field, but the computational effort with the increased hyperparameters is a significant burden.

As shown in Fig. 2, the RFI presents diverse signatures with a variety of shapes and patterns. Therefore, it is needed to extract multiscale features for accurate RFI detection. Instead of actually resampling features, the ASPP serves as a module to resample a specific feature layer before convolution at multiple scales [29]. This is equivalent to detecting the original

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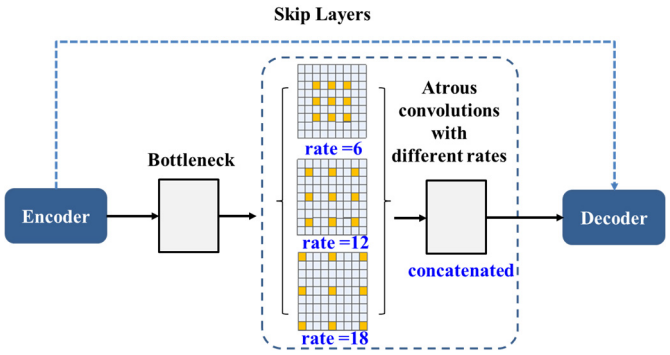


Fig. 7. Illustration of atrous convolutions with different atrous rates and its integration within the encoder–decoder architecture.

image using multiple filters with complementary effective fields of view, to capture the object and useful image context on multiple scales, as shown in Fig. 7. Different dilated rates correspond to different scopes of the receptive field, i.e., in which smaller rates could produce more detailed segmentation of the rough feature map.

Adopting this strategy can yield more context information to capture the various RFI signatures in images without increasing the number of network hyperparameters. For each pixel on the output y and filter ω, atrous convolution is applied to the input x as

$$y[i] = \sum_k x[i + r \cdot k]w[k] \tag{15}$$

where the atrous rate r determines the stride of sampling. By tuning the atrous rate, the receptive field of the convolutional filter could be varied to capture the RFI signatures with various shapes.

E. Self-Attentional Module

Sentinel-1 is a phase-preserving dual-polarization SAR system, which can transmit a signal in either horizontal (H) or vertical (V) polarization and then receive in both H and V polarizations. For conventional image processing, each channel is regarded equally without bias. However, the RFI characteristics are highly related to wave polarimetry and would have distinctive polarization signatures reflecting different polarizations with different intensities. Dual-polarization products are represented by a single composite color image in RGB with the red channel (R) representing the co-polarization VV, the green channel (G) representing the cross-polarization VH, and the blue channel (B) representing the ratio of the cross and co-polarizations [32].

In computer vision applications, each channel of the image generally shares the same background and structural information, while with different weights of the contribution to the RGB images, that is, the feature of a specific object could be detected and extracted in each channel. However, for color-coded SAR images, the situation is different. The RFI signatures are highly correlated with the polarimetric property of the emitter and receiver and thus lead to absence in certain channels. Fig. 2(a) and (b) shows the absence of RFI signatures, and this unique property would facilitate the attentional mechanism due to obvious contrast. On the other hand, the land covers also present various signatures. For example, the scattering of ocean surface behaves differently in Fig. 2(c) and (d) due to the changing of wind condition, wave direction, and so on. These two combined factors further complicated the process of RFI artifacts extraction. This explains that the focus of the attentional mechanism is different from the ones used in computer vision.

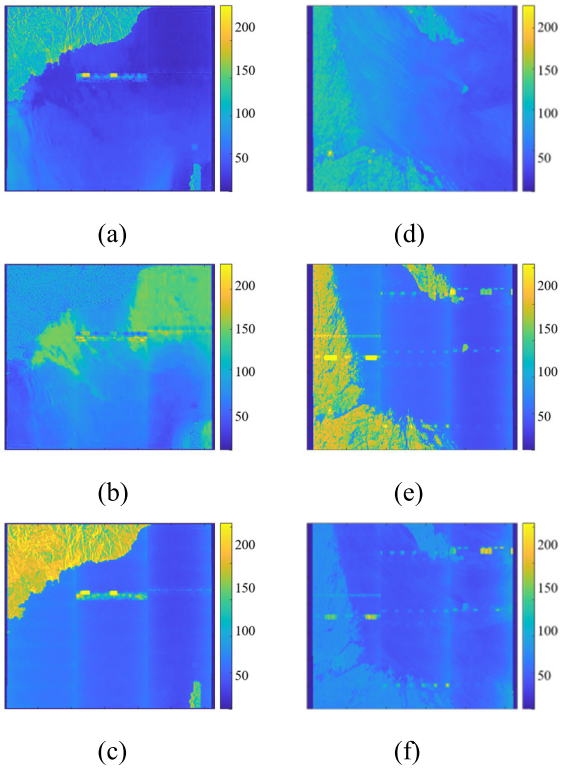


Fig. 8. Distribution of RFI signatures in different polarimetric channels. (a) and (d) VV. (b) and (e) VH. (c) and (f) |VV|/|VH|.

Fig. 8 compares the interference signatures in different polarimetric channels for two typical scenarios. It is shown that the interference energy is not distributed equally in all channels and only exists in limited spatial locations with unique patterns. For example, no obvious RFI artifacts are shown in Fig. 8(d), while many strong RFI artifacts are shown in Fig. 8(e) and (f). Therefore, it is of great importance to consider the channel difference and spatial variation for realizing the RFI detection task.

In this article, the self-attentional mechanism is proposed to incorporate channel diversity. It is composed of two parts: the channel attentional module and the spatial attentional module. The two attention mechanisms are combined in a tandem manner, as shown in Fig. 9.

The channel attentional mechanism learns the combined features of each channel and applies different weights to each channel, which is automatically calibrated by backpropagation. In this way, the interchannel relationship of features between each polarization channel is exploited, allowing the network to pay more attention to the potential salient channel.

The channel attention map is computed as [33]

$$\mathbf{M_C}(\mathbf{F}) = \sigma\left(\mathrm{MLP}(\mathrm{Avgpool}(\mathbf{F})) + \mathrm{MLP}(\mathrm{Maxpool}(\mathbf{F}))\right) \tag{16}$$

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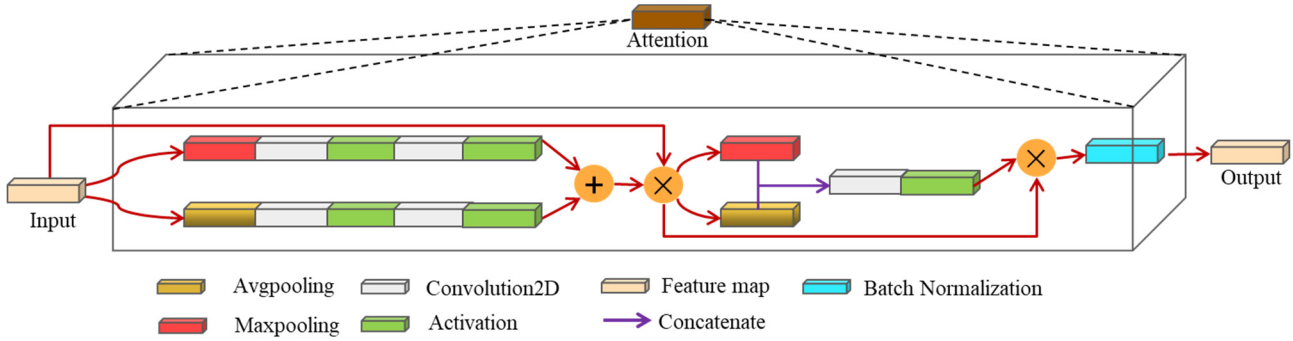


Fig. 9. Architecture of the self-attention module.

where $\mathbf{F} \in R^{C\times H\times W}$ is the input feature map.

Maxpool(·) denotes the max-pooling operation, which calculates the max value of $\mathbf{F}$ current channel. Avgpool(·) denotes the average pooling, which calculates the average pixel value of the feature map $\mathbf{F}$, which could be defined as

$$\text{Maxpool}(\mathbf{F}) = \max(\mathbf{F}(i, j)) \tag{17}$$

$$\text{Avgpool}(\mathbf{F}) = \frac{1}{W \times H}\sum_{i=1}^{W}\sum_{j=1}^{H}\mathbf{F}(i, j). \tag{18}$$

MLP(·) represents the shared multilayer perception, which consists of convolution and activation layer. It could assign the feature map of each channel a learned weight, which could be defined as

$$\text{MLP}\left(\mathbf{F}_{sq}\right) = \sum_{k=1}^{C} W_k F_{sq} + b_k. \tag{19}$$

The spatial attention module assigns different weights to all pixels on each specific channel. Like the channel attention module, the weights are automatically adjusted during the network training process based on feedback from the labels, allowing the network to focus more on the locations where interference is present while ignoring other misleading objects. The spatial attention focuses on the location of the informative part, which is complementary to the channel attention.

The spatial attention map could be calculated by

$$\mathbf{M_S}(\mathbf{F}) = \sigma\left(f^{(7\times7)}\left(\left[\text{Avgpool}(\mathbf{F}); \text{Maxpool}(\mathbf{F})\right]\right)\right) \tag{20}$$

where σ denotes the sigmoid function and $f^{(7\times7)}$ denotes a convolution operation with the filter size of 7×7.

IV. Experimental Results

A. Experimental Settings

In this section, real measured radar remote sensing images are used for performance validation and comparison. By employing the aforementioned module design, the semantic cognition capability of the network is greatly enhanced. Fig. 10 shows the overall flowchart of the proposed scheme. First, a small set of Sentinel-1 Level-1 ground range detected (GRD) images is selected for producing training datasets. Level-1 GRD products consist of focused SAR data that has been detected, multilooked, and projected to the ground range. The GRD image only contains magnitude and it has approximately square spatial resolution and square pixel spacing with reduced speckle due to the multilook processing. The RFI signatures are annotated for ground truth. Data augmentation is employed to artificially expand the size of a training set by creating modified data from the existing one. Then, the proposed semantic cognition enhanced network is trained via optimization and a well-trained model is obtained. Finally, the RFI contaminated dataset could be fed to the semantic segmentation network for RFI signature detection. Details of the experimental settings are listed as follows.

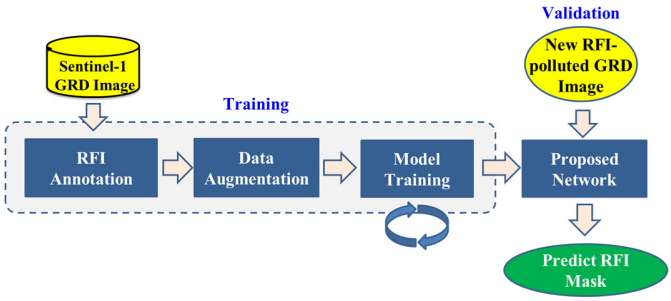


Fig. 10. Workflow of the proposed method.

1) Training and Test Data: Real measured C-band SAR images from the European Sentinel-1 observation satellite were chosen as the data source for the experiment. The training and validation datasets are 100 pairs in total, collected from five different locations worldwide containing various land covers and RFI signatures. The collected data duration is from January 2021 to June 2021. Data augmentation was applied to the training dataset to increase the diversity of the training data. In this work, we have used image rotation with the angle of 90°, 180°, and 270°, and the horizontal flip and vertical flip process to realize the data augmentation. As a result, there are a total of 600 interference images obtained for training and validation.

For performance validation, we have selected five typical scenarios all around the world from the Baltic Sea, France, Japan, Indonesia, and Central America, as shown in Fig. 11. These scenarios are representative of various land covers such as mountains, seaside, straits, and islands, as well as the inclusion of the different types of RFI signatures such

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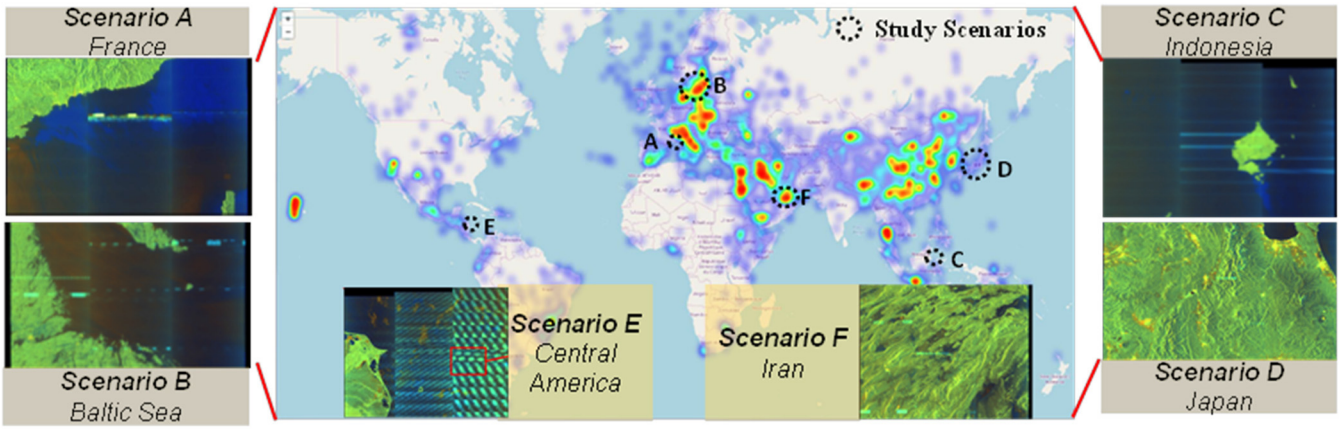


Fig. 11. Illustration of the study regions with the RFI contaminated Sentinel-1 image.

as the terrestrial RFI and intersatellite mutual interference. The RFI ground truth is manually annotated using Labelme software [36].

2) Parameter Setting and Network Training: The numbers of layers and neuron parameters of the basic network model are adopted according to [34] and [35]. A total of one independent convolution layer and 11 bottleneck layers are used, in which the convolution kernel size of the fourth, fifth, and sixth bottleneck layers is 5×5, while the size of the convolution kernel of other layers is 3×3. The initial value of the deviation of the convolution operation is set to 0, and the weight value is initialized with random uniform distribution. The activation function layer settings behind each convolution operation are different. The activation function of the last five bottleneck layers uses h-swish, while ReLU is selected for other bottleneck layers and independent volume layers. The decoder module uses two splicing operations, one $4\times$ upsampling and two $2\times$ upsampling. Compared with two $4\times$ upsampling, the convergence curve of one $4\times$ upsampling with two $2\times$ upsampling is more stable, and there is less one splicing operation compared with four $2\times$ upsampling.

We use the cross-entropy loss to evaluate the loss because the cross-entropy loss function can better control the learning rate when the sigmoid activation function is used to decrease the gradient. According to the training experience and the actual network convergency efficiency, the minibatch is set to 2, there are 25 epochs in total, and the initial learning rate is set to 0.002. When the loss of verification set under three consecutive epochs is not reduced, the learning rate will be halved, and when the learning rate is reduced for three consecutive times, the training will be terminated.

3) Compared Algorithms and Performance Evaluation Metric: For performance comparison, several SOTA techniques are used for demonstration, i.e., the Unet [25], AC-Unet [27], MobileNet2-DeepLabV3+ [37], and the proposed semantic cognitive enhancement network. For a fair comparison, all SOTA models are set with default official original parameters, such as the number of neurons, the number of network layers, and initial learning rate. The optimal results are obtained after 25 epochs of training. Each comparative method will be executed ten times in total, and the best result according to the performance evaluation metric is recorded. In this work, Cohen's Kappa coefficient, the critical success index (CSI), and the overall accuracy (OA) are utilized for quantitative evaluation.

Cohen's Kappa statistic is a metric that compares an observed accuracy with an expected accuracy and commonly used to assess the performance of a classification model. It is defined as

$$\text{Kappa} = \frac{p_o - p_e}{1 - p_e} \tag{21}$$

where p_o is the OA, which can be defined as

$$p_o = \frac{1}{N} \sum_{i=0}^{k} a_{ii} \tag{22}$$

and p_e is the measure of the agreement between the model predictions and the actual class values, which can be defined as

$$p_e = \frac{1}{N^2} \sum_{i=0}^{k} \left(\sum_{j=0}^{k} a_{ij} \times \sum_{j=0}^{k} a_{ji} \right) \tag{23}$$

where N is the total number of samples.

The CSI is defined as

$$\text{CSI} = = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} = \sum_{i=0}^{k} \frac{p_{ii}}{\sum_{j=0}^{k} p_{ij} + \sum_{j=0}^{k} p_{ji} - p_{ii}} \tag{24}$$

where the subscript i represents the ground-truth value, j represents the predicted value, $p_{(\cdot)(\cdot)}$ represents the quantity of correct and false predictions, and k represents the number of classifications. Cohen's Kappa statistic, the CSI, and the OA have a value of between 0 and 100%, where a larger value indicates a more accurate segmentation.

Table I compares the performance metric with various benchmark techniques, while the following will clarify experimental results in detail. From Table I, it is shown that the proposed method achieves higher metric scores and indicates a superior performance over the existing SOTA techniques.

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TABLE I
PERFORMANCE COMPARISON OF SEVERAL SOTA MODELS

Approaches		U-Net [25]	AC-Unet [27]	MobileNetV2-DeepLabV3+ [37]	*Proposed*
Scenario A	*Kappa*	86.94%	87.36%	87.60%	85.67%
	CSI	76.89%	78.57%	78.93%	75.47%
	OA	99.32%	99.39%	99.36%	99.31%
Scenario B	Kappa	63.79%	57.95%	61.92%	65.71%
	CSI	48.22%	42.74%	46.17%	50.37%
	OA	97.31%	96.96%	97.19%	97.52%
Scenario C	Kappa	37.25%	31.44%	43.74%	46.46%
	CSI	31.12%	27.78%	36.79%	38.40%
	OA	81.54%	79.58%	84.51%	84.06%
Scenario D	Kappa	65.85%	50.83%	56.38%	70.41%
	CSI	49.19%	34.16%	41.87%	54.44%
	OA	99.80%	99.75%	99.75%	99.83%
Scenario E	Kappa	46.54%	49.70%	47.34%	57.29%
	CSI	56.64%	53.71%	51.67%	59.63%
	OA	71.82%	75.82%	74.83%	79.34%

B. Scenario a Near France

The first scenario is observed by the European Sentinel-1A near the coastline of France on July 23, 2020, as shown in Fig. 12(a). The data acquisition orbit number is S1A_IW_GRDH_1SDV_20200723T172243_20200723T172308_033585_03E467_BC0D. Strong abnormal artifacts appeared on the sea, as well as weak RFI signatures in the right part. The artifacts are a result of typical terrestrial RFI, which is probably attributed to the radio system onboard the sea ship. The interference is a typical terrestrial RFI, which may originate from the ship's radio equipment traveling on the sea. Fig. 12(b)–(d) compares the RFI detection results after applying the Unet [25], AC-Unet [27], MobileNet2-DeepLabV3+ [37], and the proposed semantic cognitive enhancement network.

From the images in Fig. 12(b)–(f), it is shown that all the comparable methods have detected the RFI artifacts successfully. However, the Unet and AC-Unet are biased to reflect the rectangular shape in the ground-truth label. MobileNet2-DeepLabV3+ and the proposed semantic cognitive enhancement network have a more smooth boundary that better characterizes the patterns of the RFI signature. Since it is a simple scenario with RFI artifacts appearing on the sea, the metrics of the proposed MobileNet2-DeepLabV3+ achieve the highest value.

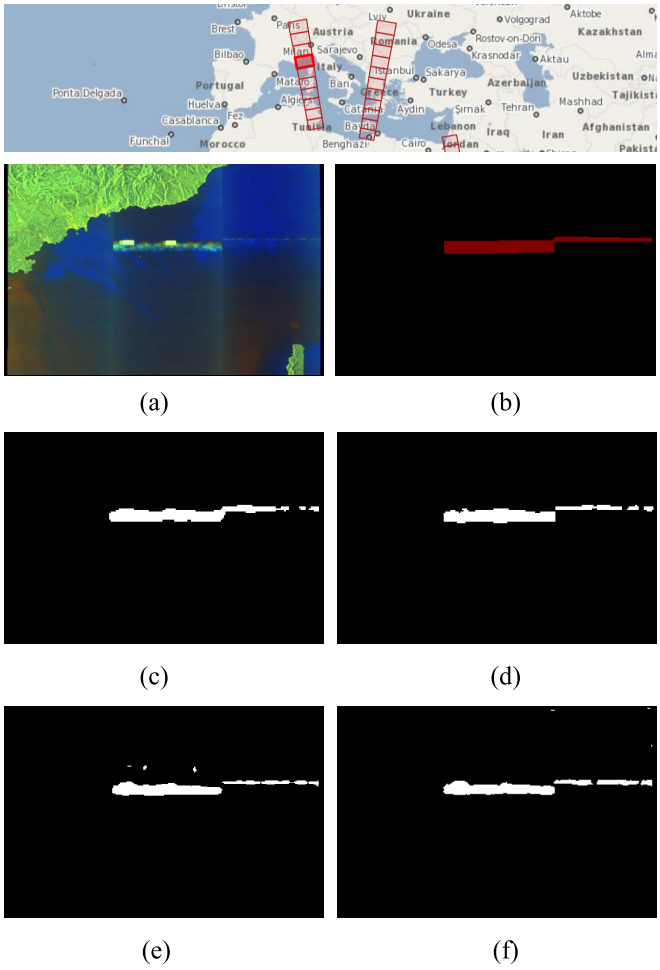


Fig. 12. Comparison of RFI detection performance for scenario A: (a) original RFI-polluted image, (b) reference ground-truth label, (c) U-Net, (d) AC-Unet, (e) MobileNetV2-DeepLabV3+, and (f) proposed semantic cognitive enhancement net.

C. Scenario B in Baltic Sea

This section demonstrates the results acquired in the Baltic Sea near Denmark on June 1, 2021, as shown in Fig. 13(a). The data acquisition orbit number is S1A_IW_GRDH_1SDV_20210601T162920_20210601T162945_038149_0480A6_5057. The RFI artifacts are much more complicated with various spatial distributions and radiometric textures, encompassing both strong and weak RFI blocks. The illuminated Baltic Sea area is a very busy international shipping lane, and thus, the RFI signatures are a result of radiolocation equipment from multiple ships.

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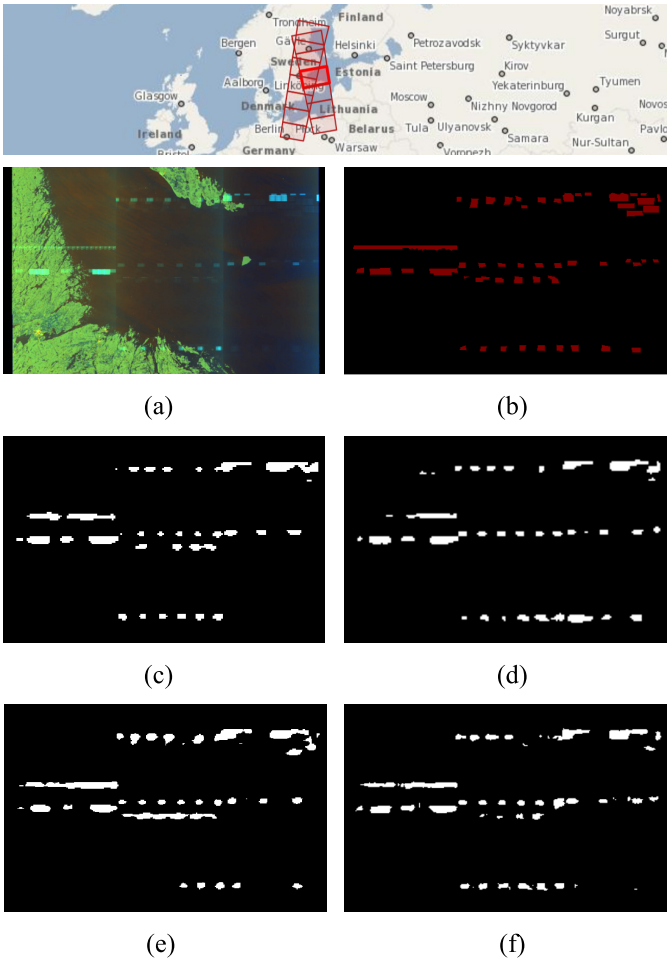


Fig. 13. Comparison of RFI detection performance for scenario B: (a) original RFI-polluted image, (b) reference ground-truth label, (c) U-Net, (d) AC-Unet, (e) MobileNetV2-DeepLabV3+, and (f) proposed semantic cognitive enhancement net.

The segmentation results in Fig. 13(b)–(f) show that the Unet can accurately identify all interference with high energy, but weak target interference below is missed more often. The AC-Unet network identifies all of the lower interference targets but misses all of the weak energy interference in the middle and fails to identify the "raindrops" on the left side that overlap with the land. The proposed method does not miss the strong interference targets and has a better detection for weak RFI artifacts than U-net. Although some of the weak targets have a smaller segmentation area, it achieves the highest value of evaluation metrics.

D. Scenario C in Indonesia

Scenario C is a typical image polluted by interstellar mutual interference with varying energy levels, which is acquired by Sentinel-1A in the Indonesia island on January 4, 2021. The data acquisition orbit number is S1A_IW_GRDH_1SDV_20210104T223058_20210104T223127_035994_043783_E620. The regular stripes in Fig. 14(a) are a product of mutual interference from other microwave satellites operating in adjacent spectrum and areas. The long-duration mutual RFI exists in most of the bursts, which undergoes partial coherent processing and produces bright stripes with high intensity.

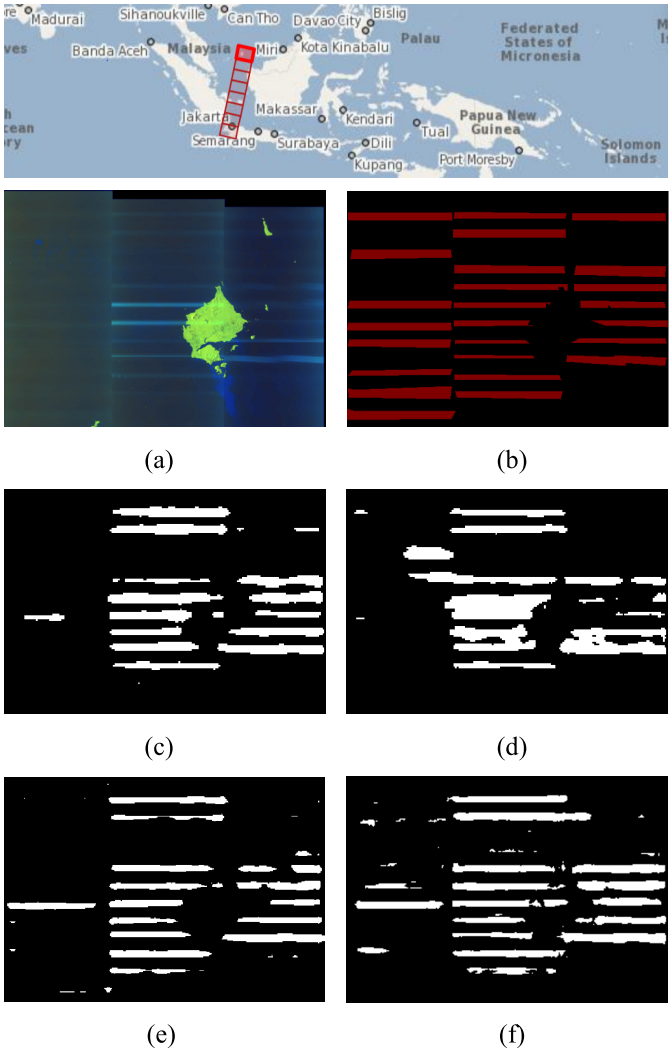


Fig. 14. Comparison of RFI detection performance for scenario C: (a) original RFI-polluted image, (b) reference ground-truth label, (c) U-Net, (d) AC-Unet, (e) MobileNetV2-DeepLabV3+, and (f) proposed semantic cognitive enhancement net.

The segmentation results in Fig. 14(b)–(f) show that the comparison techniques can extract the features of such interference with equally spaced occurrences and strip shapes, with AC-Unet having a clear false detection that some stripes are merged. The proposed method achieves superior performance with works best with fewer missed detections of weak energy RFI artifacts, which could be verified from the calculation of the value of the metrics.

E. Scenario D in Japan

To further illustrate the performance on weak RFI artifacts, we present Scenario D acquired in Japan in July 2021. The data acquisition orbit number is S1A_IW_GRDH_1SDV_20210226T205135_20210226T205200_036766_045283_FD11. Unlike the previous types of interference better isolated in the sea area, the two disturbances in this scene are all located in the land region, as shown in Fig. 15(a) and (b). Since the energy at ground level

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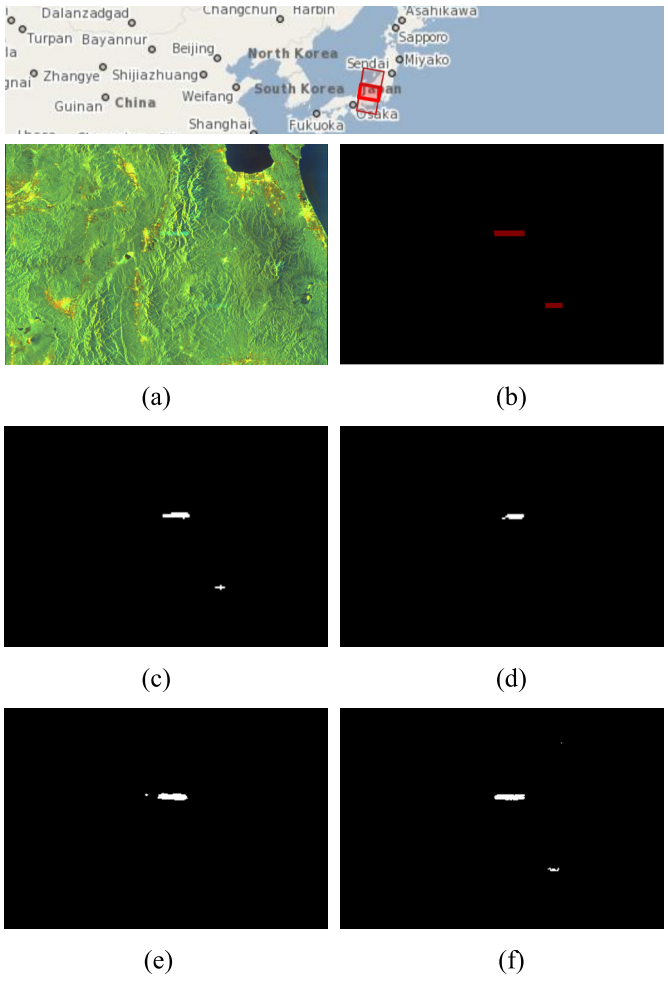


Fig. 15. Comparison of RFI detection performance for scenario D: (a) original RFI-polluted image, (b) reference ground-truth label, (c) U-Net, (d) AC-Unet, (e) MobileNetV2-DeepLabV3+, and (f) proposed semantic cognitive enhancement net.

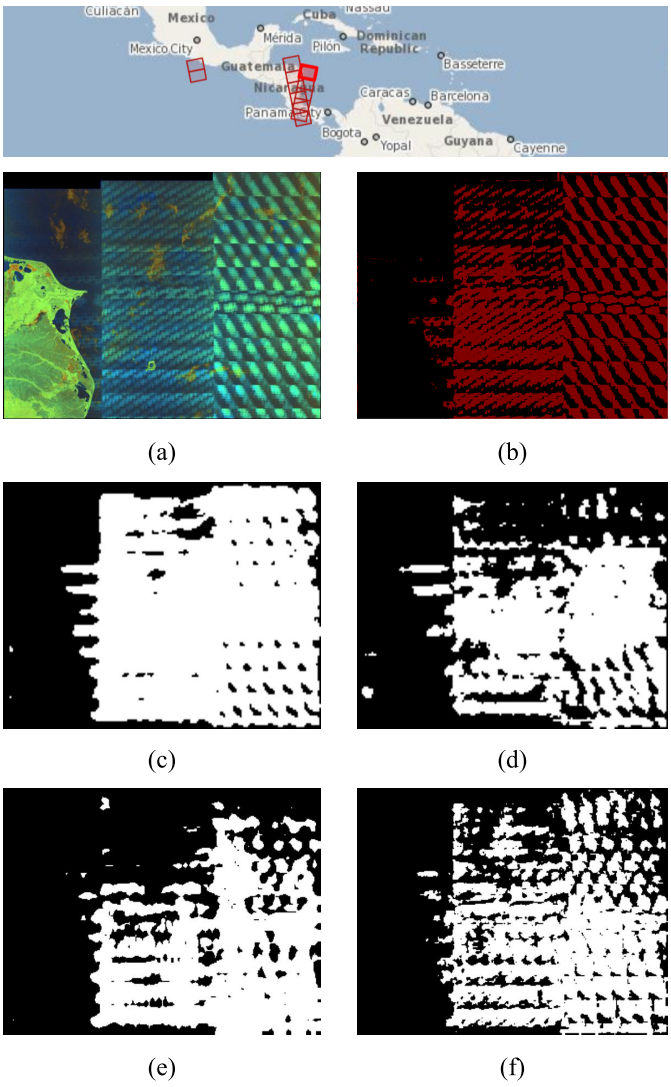


Fig. 16. Comparison of RFI detection performance for scenario E: (a) original RFI-polluted image, (b) reference ground-truth label, (c) U-Net, (d) AC-Unet, (e) MobileNetV2-DeepLabV3+, and (f) proposed semantic cognitive enhancement net.

is significantly higher than at sea level, the RFI artifacts present rather similar radiometric characteristics with the land covers.

From the segmentation results plotted in Fig. 15(c)–(f), the Unet and the proposed method can identify the interference in the bottom-right corner, while the AC-Unet and the MobileNetV2-DeepLabV3+ have missed detection. The larger metric values indicate that the proposed semantic cognition enhanced network has a better performance when dealing with the weak RFI components.

F. Scenario E in Central America

In this case, a very complicated RFI scene is presented, which is acquired by Sentinel-1A in Central America on October 31, 2017. The data acquisition orbit number is S1A_IW_GRDH_1SDV_20171031T112139_20171031T112204_019056_0203AE_0646. To test the generalization performance, it is worth noting that the RFI artifacts in this scenario are not considered for training. As shown in Fig. 16(a) and (b), the RFI case in this scenario is a typical space-to-space mutual interference, presenting strong RFI signatures with very unique textures. To better verify the generalization of the model, this type of interference is not included in the training. Fig. 16(c)–(f) shows the comparison results. The neural networks could segment the disturbance from the background to some extent, but the texture is not segmented as well as the proposed method. AC-Unet and MobilenetV2-DeeplabV3+ have a lot of miss detections in the upper part, while Unet merged the majority part of the right side as a whole. The proposed network is better at characterizing the graininess of the interference signature and extracting out the unique texture patterns corresponding to the space mutual interference.

G. Ablation Study and Discussion

The ablation study studies the performance of a neural network by removing certain components, to understand the

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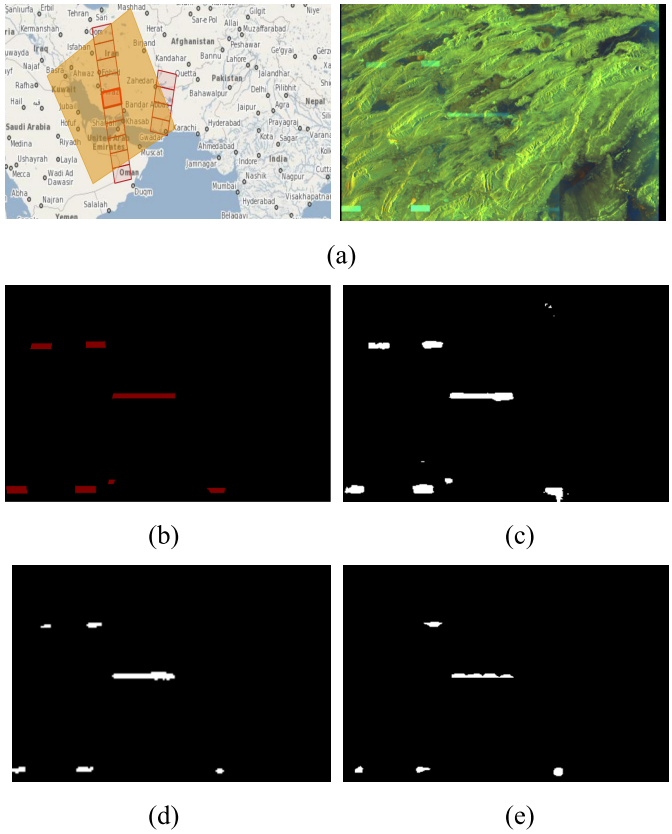


Fig. 17. Comparison of RFI detection performance for scenario F: (a) original RFI-polluted image, (b) reference ground-truth label, (c) proposed, (d) no ASPP, and (e) no self-attention.

TABLE II
ABLATION STUDIES

Approaches	*Kappa*	*CSI*	*OA*
No ASPP	72.15%	56.80%	99.53%
No Self-attention	59.22%	42.42%	99.19%
Proposed	84.37%	73.31%	99.66%

contribution of the component to the overall system [38]. The key modules of the proposed network are the ASPP, depthwise convolution, and self-attentional mechanism. Since the depthwise convolution is mainly designed for reducing the computation burden and plays little role in the accuracy improvement. Therefore, in this article, we perform the ablation studies for the ASPP, depthwise convolution, and self-attentional mechanism. The testing scenario is acquired in the Iran region on February 27, 2021. The data acquisition orbit number is S1A_IW_GRDH_1SDV_20210227T142546_20210227T142611_036777_0452E0_D0AA. Fig. 17 compares the segmentation results. It is shown that the proposed network better captures the weak RFI components, while miss detections may happen for the lack of ASPP or self-attention modules. Table II summarizes the Kappa statistic, CSI values, and OA for different network architectures, which further verifies the functionality of the key modules.

V. CONCLUSION

Active remote sensing system such as Sentinel-1 faces a great challenge to accurate observation due to the prevalence of RFI. Most existing works mainly perform interference detection on raw data or detect the existence of RFI for a single image without specifying the exact part. In this article, an alternative RFI signature detection and extraction scheme is proposed using a semantic cognitive enhancement network, which is introduced. The superiority of the proposed methods lies in the following aspects.

1) The proposed semantic cognition network adopts an encoder-to-decoder structure, which enables the flagging of RFI in a pixelwise manner. The semantic cognition capability is enhanced by incorporating the design of ASPP, depthwise convolution, and self-attentional mechanism.

2) Extensive experimental results on Sentinel-1 images observed from five different scenarios with various RFI types, including the common terrestrial RFI and complicated satellite mutual interference, are demonstrated. The proposed method does not require a large volume of training image pairs as well as prior accurate labeling and has good generalization performance on datasets outside of the training dataset. Experiments show promising results on both strong and weak interference under the condition of small samples.

3) The proposed method can be used as a supplement to the existing approaches and is very suitable for the integration of mass image processing. It is interesting to monitor the evolutions of occurrences and intensity, which allows deriving temporal and geographic patterns of such future potential RFI. The output RFI flagging results are beneficial for inferring radiation sources and RFI case reporting.

The data-oriented method is a good attempt to improve the performance of RFI detection. In the following, we would like to explore a more refined network by incorporating model prior knowledge and further improving the accuracy in terms of dealing with the complicated mutual interference signatures.

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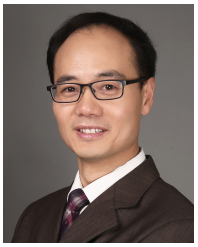
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